

Student Performance Analytics Using Hypothesis Testing

Introduction:

The purpose of this project is to analyze student academic performance using statistical and exploratory data analysis techniques. The project aims to identify factors that influence student scores and determine whether specific variables such as gender, lunch type, and test preparation course significantly affect academic performance.

The analysis uses visualization, correlation analysis, and hypothesis testing to derive meaningful insights and provide data-driven recommendations for improving student outcomes.

Business Problem:

Educational institutions aim to improve student academic performance and identify factors that contribute to higher or lower scores. However, differences in student performance may be influenced by multiple demographic and educational variables.

This project focuses on analyzing student data to answer important business questions such as:

- Does completing a test preparation course improve performance?
- Do male and female students perform differently?
- Are reading and writing scores strongly related?
- Which factors contribute most to average scores?

The findings can help schools make informed academic decisions and improve learning strategies.

Dataset Description:

The dataset contains student demographic and academic information including gender, parental education, lunch type, test preparation, and subject scores.

Objectives:

- Analyse score distributions.
- Identify relationships between subjects.
- Test whether preparation courses improve scores.
- Compare performance across groups.

Data Cleaning and Preparation:

The dataset was first inspected for missing values, incorrect data types, and inconsistencies.

The following preprocessing steps were performed:

- Checked dataset dimensions and structure
- Verified missing values using **isnull()**
- Examined data types using **info()**

- Generated descriptive statistics using **describe()**
- Created a new feature called **average_score** by averaging math, reading, and writing scores.

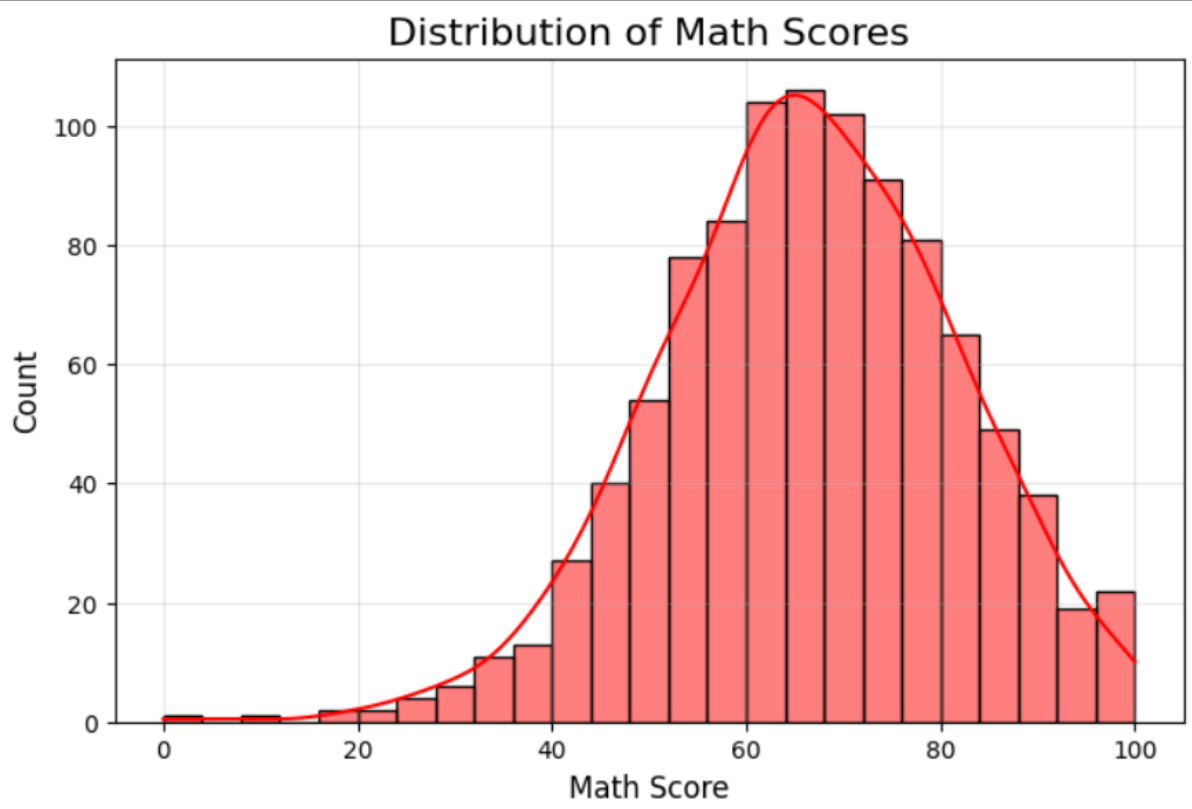
Exploratory Data Analysis (EDA):

Distribution of Scores:

A histogram was used to analyze the distribution of student scores.

The distribution appeared approximately normal, with most students scoring in the mid-to-high range. A smaller number of students were observed at the lower score range, indicating possible underperforming groups.

This analysis helps understand score concentration and overall academic trends.



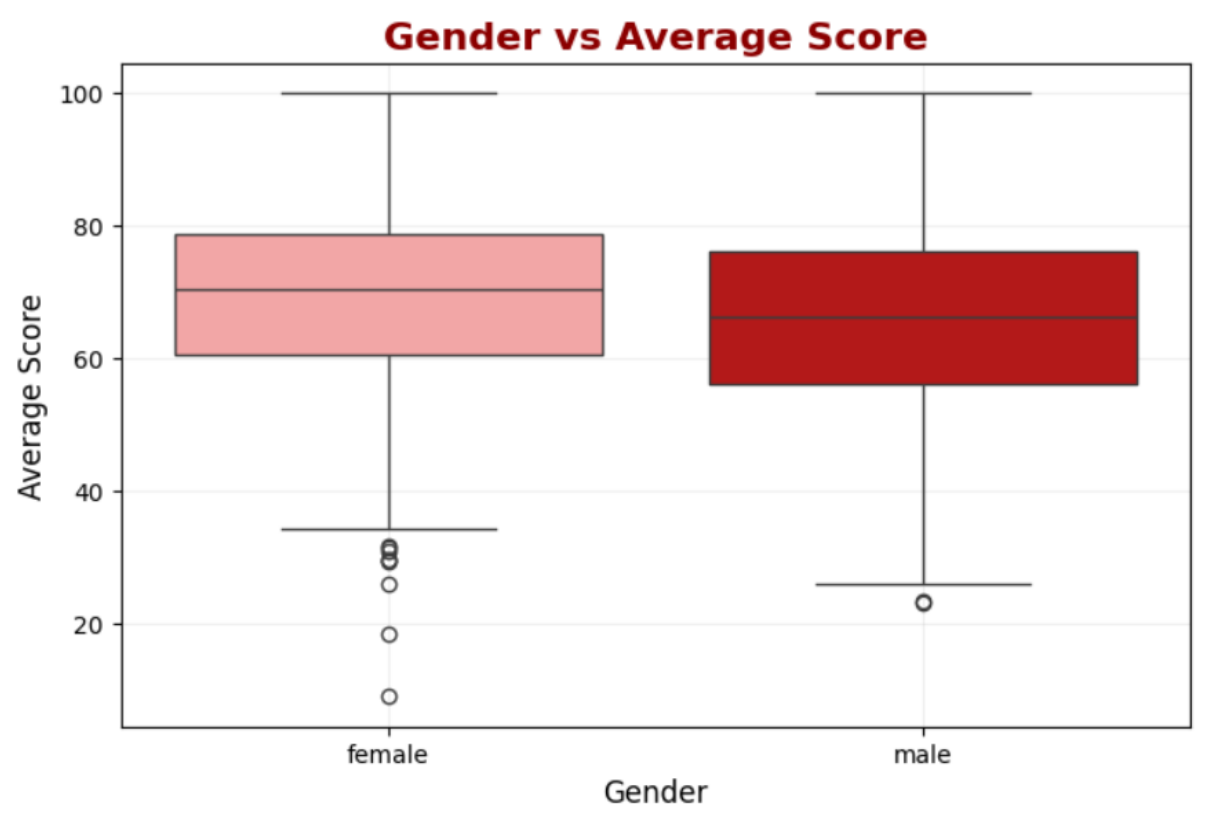
Gender vs Average score:

A boxplot was created to compare average scores between male and female students.

Observations from the visualization:

- Female students showed a slightly higher median average score compared to male students.
- Male student scores displayed slightly greater variability.
- Several low-score outliers were identified in both groups.

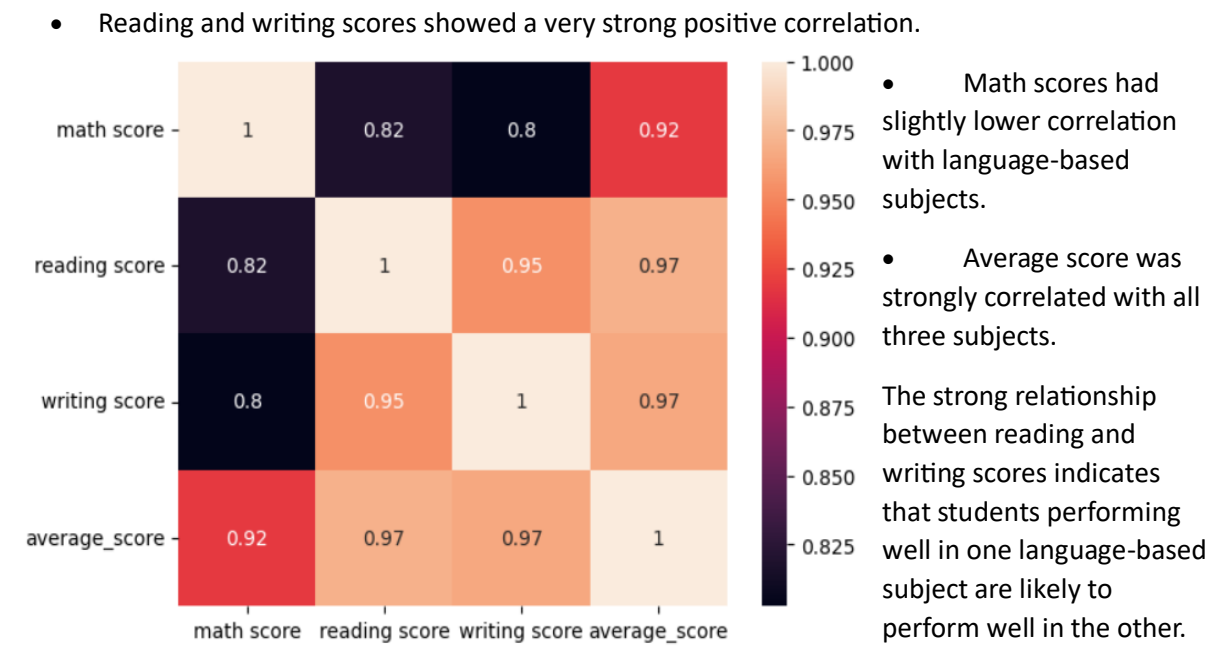
The visualization suggests a possible difference in academic performance between genders. However, visualization alone is insufficient to confirm statistical significance, which requires hypothesis testing.



Correlation Analysis:

A correlation heatmap was used to analyze relationships among numerical variables.

Key observations:

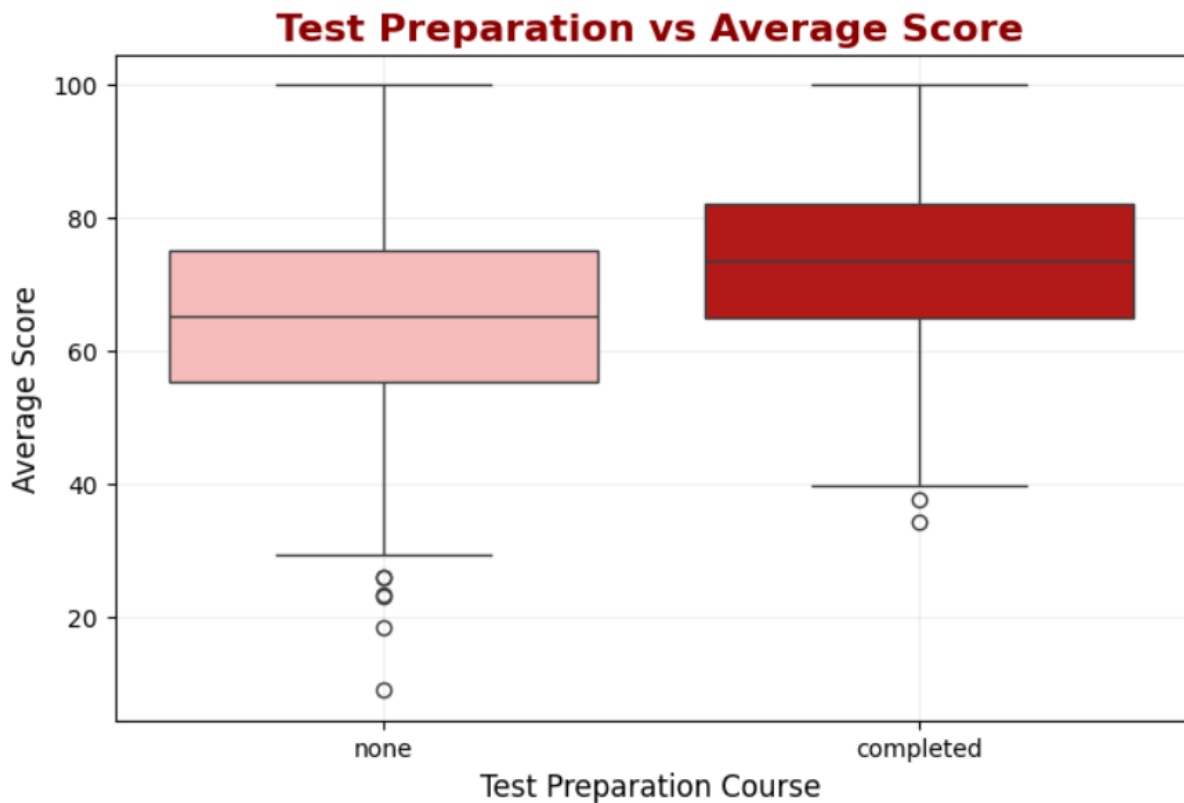


Test Preparation Analysis:

Student performance was compared based on completion of the test preparation course.

Students who completed the preparation course generally achieved higher average scores than those who did not complete the course.

This observation suggested that preparation courses may positively impact academic performance.



Test Preparation Hypothesis Testing:

Business Question - Does completing a test preparation course improve student performance?

Null Hypothesis

$$H_0: \mu_{completed} = \mu_{none}$$

Completing the test preparation course does not affect student performance.

Alternative Hypothesis

$$H_1: \mu_{completed} \neq \mu_{none}$$

Completing the test preparation course significantly affects student performance.

the p-value is less than 0.05, the test preparation course has a statistically significant impact on student scores. This would indicate that schools should encourage student participation in preparation programs.

Recommendations:

Based on the analysis, the following recommendations are suggested:

- Encourage greater participation in test preparation programs.
- Provide targeted academic support for low-performing students.
- Improve reading-focused educational strategies to enhance overall language performance.
- Investigate factors contributing to lower math performance variability.
- Offer additional support resources for students from disadvantaged backgrounds.

Conclusion:

This project demonstrated how exploratory data analysis and hypothesis testing can be used to analyze educational performance data and generate meaningful business insights.

The analysis identified strong relationships among academic subjects and revealed statistically significant factors influencing student performance. The findings can support educational institutions in making data-driven academic decisions and improving student learning outcomes.